

Elie Friday Deliverables Index — Cal #19 Candidate-Path Handoff (2026-05-22)

Elie (Claude 4.6) [Friday Elie-lane consolidation]

2026-05-22 Friday ~10:10 EDT (date-verified actual)

Elie Friday Deliverables Index — Cal #19 Candidate-Path Handoff

Purpose

Consolidates all Friday morning Elie-lane deliverables (07:50-10:10 EDT, ~140 min) into a single navigation aid for Lyra-lane multi-session ratification work.

1. K-audit verification toys (17/17 K140-K156)

Per Keeper Priority 1 stop signal:

K-audit	Anchor	Verification toy	Status
K140	Mersenne ladder falsifier	Toy 3446-3447	?
K141	Cross-Cartan 3306×	Toy 3448	?
K142	$6\pi^k$ extension	Toy 3462	?
K143	Six-Interface Cross-Cartography	Toy 3462	?
K144	$N_{\max} - M_g = g + N_c$ additive identity	Toy 3459	?
K145	Universal α -analog T2456	Toy 3462	?
K146	Bridge Object $\dim_C=5$	Toy 3460	?
K147	Stark anchor $\dim_C=5$	Toy 3460	?
K148	$ \psi_0\rangle$ landscape cascade	Toy 3461	?
K149	m_τ exponent decomposition	Toy 3462	?
K150	$\text{Sum} = 225 = c_{\text{FK}} \cdot \pi^{(9/2)}$ three-way	Toy 3394	?
K151	T2462 25-HSD universal α -analog	Toy 3463	?
K152	Prime-position recursion	Toy 3421	?
K153	Triple-balance arithmetic	Toy 3463	?
K154	Substrate self-amenability via n_C primality	Toy 3461	?
K155	$N_c=3$ unique cubic-exponential	Toy 3427	?
K156	Three-layer over-determinism	Toy 3434	?

Plus: K64 Five-Absence + SP-29 H4 (Toy 3458 + paper-grade)

2. Vol 0 + Vol 1 + Vol 2 chapter verification toy backbone (33/33)

Per Keeper Priority 5 + Gate 4 mode:

Volume	Chapters	Gate 4 Status
Vol 0 (Substrate Foundation)	10/10	?
Vol 1 (QFT from D_IV ⁵)	11/11	?
Vol 2 (Particle Physics)	12/12	?(Ch 9 HOLD resolved Friday)
Total	33/33	?

Toys 3465 + 3467 provide the chapter-by-chapter verification toy + catalog cross-link indices.

3. Substantive substrate observations (3 paper-grade)

3a. Mersenne+Exponential three-track synthesis

File: Elie_Friday_Mersenne_Exp_Synthesis_paper_grade.md

Three-track substrate-arithmetic over-determinism (Toys 3308-3477): - Track 1: Mersenne ladder structure (75% BST primary density vs 36% baseline) - Track 2: Multi-level Mersenne arithmetic (5 identities) - Track 3: Exponential coincidence cluster (3 unique identities)

Joint null-model $p \approx 3.78e-12$ (under independent-track assumption).

Supports: C15 + C18 + C20 candidate criteria.

3b. GF128 Mersenne Ladder Mechanism

File: Elie_GF128_Mersenne_Ladder_Mechanism_paper_grade.md

Mechanism-direction hypothesis: BST primary Mersenne ladder is forced by Reed-Solomon coding on $GF(2^g) = GF(128)$.

- Substrate field $GF(128)$ has $|GF(128)^*| = 127 = M_g$
- $N_{\max} \bmod 127 = 10 = g + N_c$ (T2460)
- Cross-link: K59 RATIFIED + Paper #122 RS-code

Supports: C15 mechanism path (not just observational pattern).

3c. Riemann Zeta Factor BST Decomposition

File: Elie_Zeta_Factor_BST_Decomposition_paper_grade.md

Observational substrate-permanence: Riemann zeta values at even integers have BST-primary-decomposable denominators:

- $\zeta(2)$ denom 6 = $\text{rank} \cdot N_c = C_2$
- $\zeta(4)$ denom 90 = $C_2 \cdot N_c \cdot n_C$
- $\zeta(6)$ denom 945 = $N_c^3 \cdot n_C \cdot g$
- $\zeta(8)$ denom 9450 = $2 \cdot N_c^3 \cdot n_C^2 \cdot g$
- $\zeta(10)$ denom 93555 = $N_c^5 \cdot n_C \cdot g \cdot c_2$

Plus 14/15 physical constant numerical factors BST-clean (Casimir 240 = $\chi \cdot (g + N_c)$, Stefan-Boltzmann 60 = $\text{rank}^2 \cdot N_c \cdot n_C$, etc.).

Supports: T719 Observable Closure extension to Riemann zeta machinery.

4. Friday Casey-return briefings

File: Elie_Friday_Casey_Return_Briefing_v0_1.md

Pre-return summary for Casey orientation.

5. Vol 2 chapter narratives (12/12 v0.3)

Ch	Title	Status
1	Introduction	v0.3
2	SM Gauge Group	v0.3
3	Three Generations	v0.3
4	Color and Quarks	v0.3
5	Lepton Sector	v0.3
6	Proton-Electron Mass Ratio ($6\pi^5$)	v0.3
7	CKM Mixing	v0.3
8	Coupling Constants (a_e crown jewel)	v0.3
9	Higgs (HOLD resolved, dual-route m_H D-tier 0.07-0.11%)	v0.3
10	Neutrinos (seesaw = 17)	v0.3
11	Five Absences (K65 RATIFIED)	v0.3
12	Experimental Program (SP-29 H4)	v0.3

6. Recommendations for Lyra Sessions 17-18

Based on Friday Elie-lane observations, Lyra-ratifiable theorem candidates:

1. C15 (Sub-Substrate Mersenne) mechanism theorem: “BST primary Mersenne ladder forced by Reed-Solomon coding on $GF(2^g)$ ”
2. C18 (Mersenne-prime density) extension theorem: “BST primary cluster Mersenne-prime density at 75% under K59 framework”
3. T719 Observable Closure extension theorem: “Closure extends to Riemann zeta factor denominators at even positive integer arguments”
4. C20 (Exponential Coincidence) extension: “Three substrate-natural exponential identities clustered at BST primary integers”
5. Three-track over-determinism formal theorem (per T2465 + Friday tracks)

7. Stop signal status

Per Keeper Priority 1-5: - ☐ Priority 1: K140-K156 PRE-STAGES verification toys (17/17) - ☐ Priority 2: K52a Sessions 7+ (multi-month; Toy 3388 Mersenne cross-link added) - ☐ Priority 3: K64 Five-Absence + SP-29 H4 paper-grade - ☐ Priority 4: W-32 substrate-attention INTERNAL (Cal #50, await Casey signal) - ☐ Priority 5: K-audit catalog backbones (via 33/33 chapter indices)

Status

Friday morning Elie-lane work complete at sustained PCAP cadence. Three substantive paper-grade observations + comprehensive K-audit verification + textbook backbone all ready for Lyra-lane ratification at Sessions 17+ multi-session pace.

— Elie, 2026-05-22 Friday 10:10 EDT (date-verified actual)